

TRAILBLAZE

INTRODUCTION

Trailblaze is an autonomous robotic challenge where bots are designed to navigate a predefined path using sensors to detect and follow a black line. This event tests the efficiency, speed, and intelligence of the robotic systems developed by participants. The competition encourages innovation, engineering prowess, and teamwork to build the most precise and swift autonomous robot. Can your bot outmanoeuvre the rest and emerge victorious in this high-tech showdown?

PROBLEM STATEMENT

Design and build an autonomous robot that can detect and follow a black line on a contrasting surface. The robot must complete the track in the shortest possible time while maintaining accuracy.

Deviations from the path or failure to complete the track will result in penalties.

ROBOT SPECIFICATIONS

- The robot must be fully autonomous and must not require any human intervention during the run.
- Maximum robot dimensions: 25cm x 25cm x 25cm (LxBxH).
- The robot must use sensors (such as IR or camera-based) to detect the line.
- The robot may be battery-operated; the total voltage must not exceed 13V DC at any point.
- The use of pre-built kits are not allowed; although the individual parts such as chassis and wheels are allowed; custom-built designs are encouraged.

ARENA SPECIFICATIONS

- The black line width will be 2 cm, placed on a white surface.
- The track may include sharp turns, curves, and intersections.
- There will be no AC supply from our side.

GAMEPLAY

- The robot must autonomously follow the black line without any manual control.
- There will be 2 stages:
- **Stage 1:** being the qualifying round (easy map would be provided).
- **Stage 2:** being the final round (a more complicated map would be provided)
- Those teams qualifying the stage 1 will only be eligible to participate in stage 2.
- The robot exceeding the size limits would be heavily penalized and may be disqualified solely based on the decisions of judges.
- The robot must stay within the track; veering off will lead to penalties.
- The organizers reserve the right to disqualify any team due to safety concerns or rule violations.
- Participants must present valid student identity cards.
- If a robot is unable to move or follow the track correctly, it will be disqualified from that round.
- In case of a draw (in terms of points), the robot with least time to complete the track will be declared the winner.

EVALUATION CRITERIA

The robot that completes the track in the shortest time with the least penalties will be declared the winner.

GUIDELINES

- **Eligibility:** Students pursuing Undergraduate Degrees/ Master's Degrees in any discipline.
- **Team Size:** A team can have a minimum of 3 and a maximum of 5 members.
- **No Double Troubles:** Only one entry would be acceptable. In case of multiple entries, the latest one would be considered for evaluation.

QUERIES :

For any queries, you can contact:

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Anuradha Verma | 9660347336

OR

Just mail us with the Subject: "**Query | EVENT NAME | TeamName**" at : ju.makerspace@jecrcu.edu.in